

Cascading Failure in Distributed Systems

Midterm Mastery - II

Architecture



- Search makes synchronous call to Inventory
- Inventory intentionally slow + unreliable
- Load: 100 concurrent users
- Tight synchronous coupling between Search and Inventory.

Problematic Deployment

Inventory:

- Latency up to 3000ms
- 30% failure rate

Search:

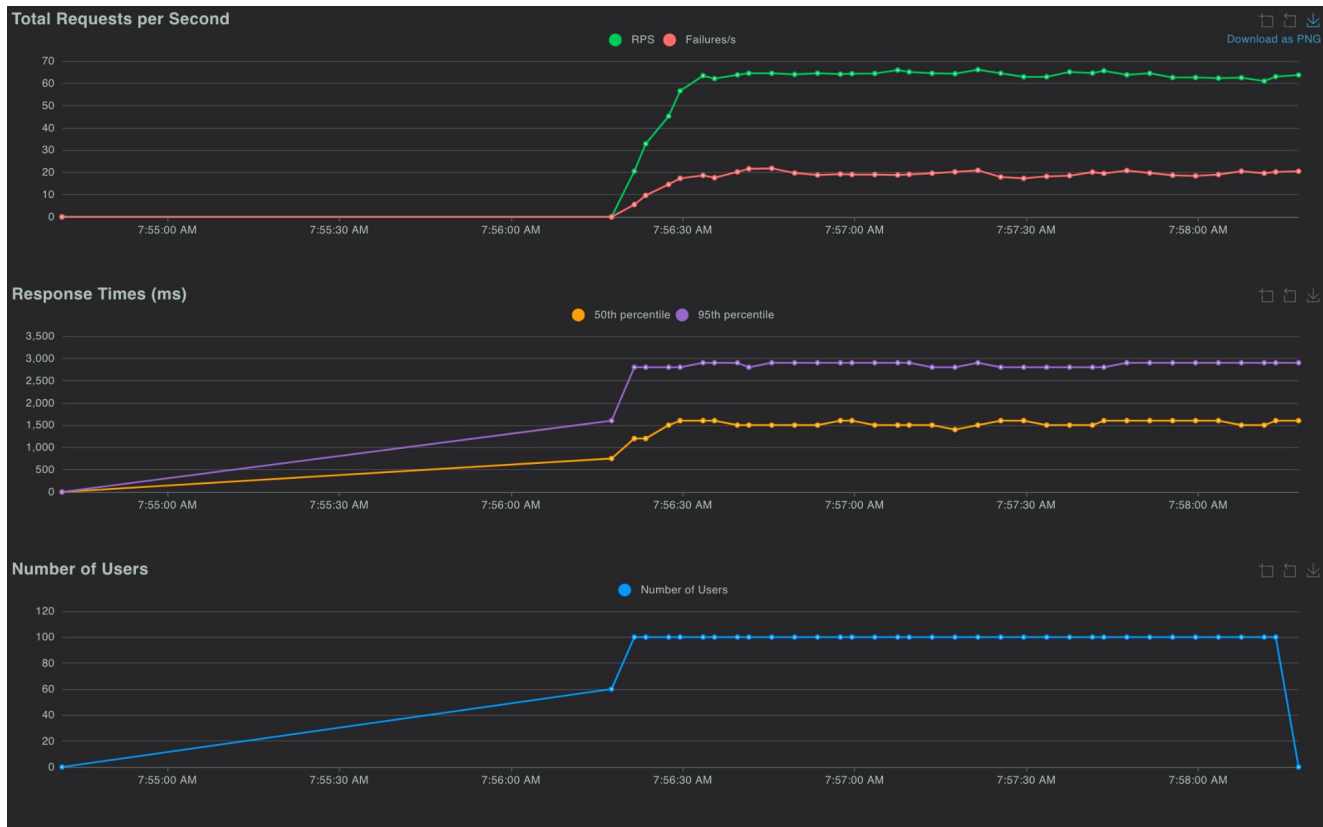
- No timeout
- Blocking downstream call

Results:

- $p95 \approx 3000\text{ms}$
- 15–20 failures/sec
- ~60 RPS

Under load, Search blocked on slow Inventory responses.

Locust Results



Root Cause

No timeout on downstream dependency.

- Slow Dependency
- Thread blocking
- Request buildup
- Latency amplification
- 5xx errors

Resilience Strategy

Fail-Fast Pattern

- 200ms timeout
- Fallback response
- Prevent blocking

Code snippet: *context.WithTimeout(200 * time.Millisecond)*

Bounded downstream wait time to 200ms.

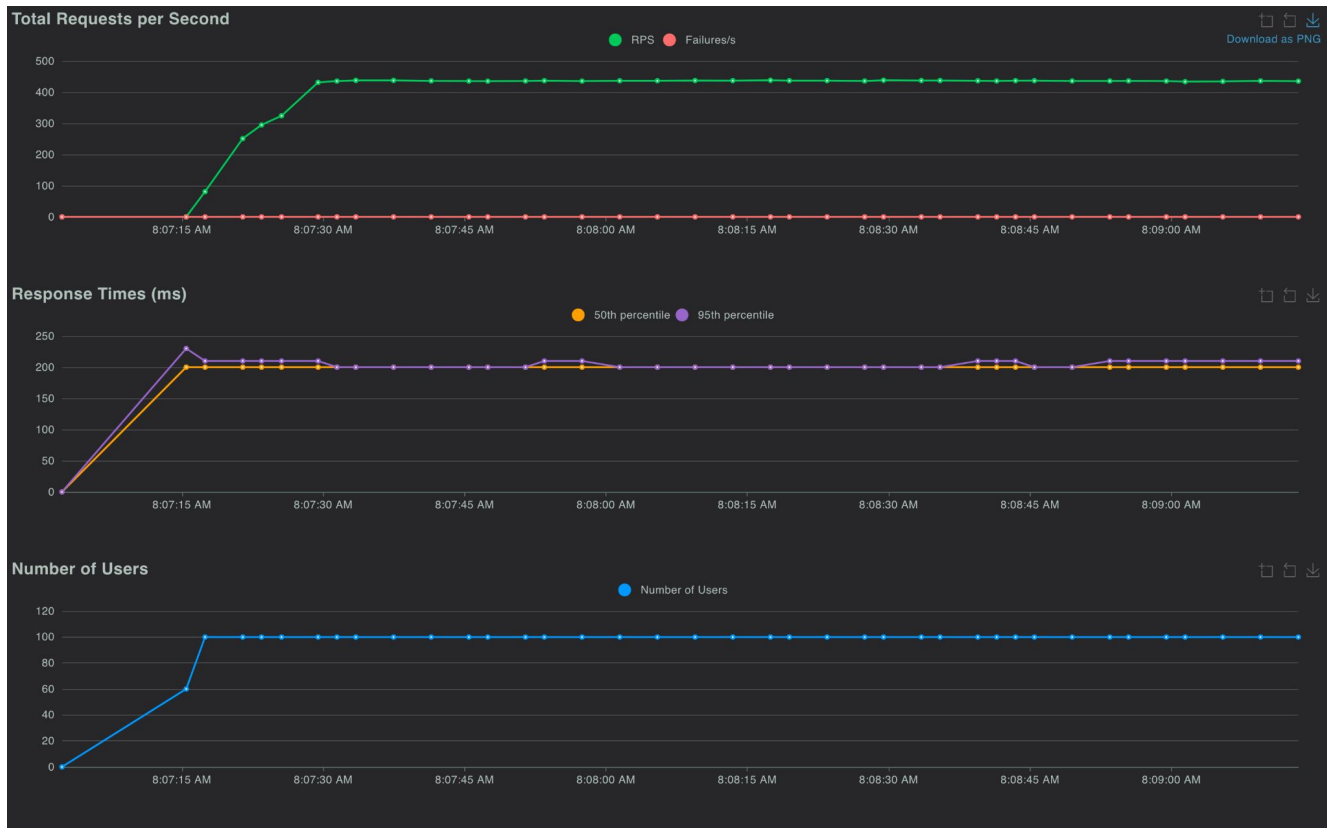
Improved Results

Fail-Fast Pattern

- Same load (100 users)
- $p_{95} \approx 200\text{ms}$
- Failures ≈ 0
- 430+ RPS

Latency bounded and system stabilized under identical load.

Improved Locust Results



Thank You!